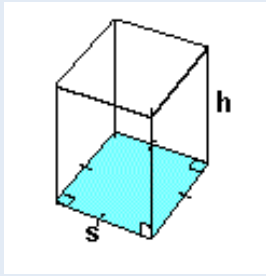


Part 2: Minimize the Surface Area of a Square Based Prism

Recall:



$$\text{Volume of a Prism} = (\text{Area of Base})(\text{Height})$$

$$\text{Volume of Square Based Prism} = (s^2)(h)$$

$$\text{Surface Area of a Square Based Prism} = 2s^2 + 4sh$$

Minimizing surface area for a given volume is important when designing packages and containers to save on materials and reduce heat loss.

Investigation: You will determine how to find the minimum surface area of a square based prism for a given volume.

a) Complete the following table to determine the dimensions of the square based prism that has a minimum surface area for a fixed volume of 64 m^3

Note: You can calculate height given the volume and side length using $h = \frac{\text{volume}}{\text{length}^2}$

	Side Length of Square Base (s)	Area of Square base (s^2)	Height (h)	Volume (s^2h)	Surface Area ($2s^2 + 4sh$)
1	1	1	64	64	258
2	2			64	
3	3			64	
4	4			64	
5	5			64	
6	6			64	
7	7			64	

b) Which prism has the least surface area?

c) Describe the shape of this prism

Results:

For a square based prism with a given volume, the minimum surface area occurs when the prism is a _____.

Given a volume, you can find the dimensions of a square-based prism with minimum surface area by solving for s in the formula $V = s^3$. This is the formula for the volume of a cube where V is the given volume and s is the length of a side of the cube. You can solve for the side length by taking the cubed root of the volume.

For example: The cubed root of 64 = $\sqrt[3]{64} = 64^{1/3} = 4$

Apply What You Learned

1) a) What would be the dimensions of the square-based prism with a minimum surface area given a volume of 27 cubic units?

b) Calculate the surface area (*Surface area of a cube = $6s^2$*)

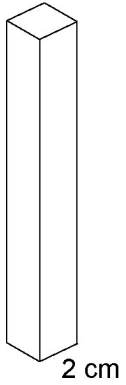
2) Talia is shipping USB cables in a small cardboard square-based prism box. The box must have a capacity of 750cm^3 and Talia wants to use the minimum amount of cardboard when she ships the box.

a) What should the dimensions of the box be, to the nearest hundredth of a centimeter?

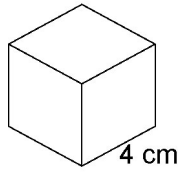
b) What is the minimum amount of cardboard that Talia will need, to the nearest tenth of a square centimeter? (*Surface area of a cube = $6s^2$*)

3) Each of these square-based prisms has volume 64 cm^3 .

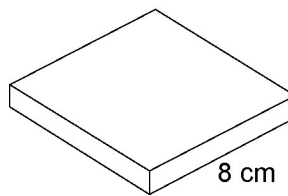
Prism A



Prism B



Prism C



a) Find the height of each prism.

b) Find the surface area of each prism.

c) Order the prisms from least to greatest surface area.